

SMPD2 (O60906) — Adjudication of caveola (GO:0005901) and response to mechanical stimulus (GO:0009612) annotations

Focus: function_assignment · hypothesis slug `caveola-mechanosignaling-antigen-identity`
Gene: human SMPD2 / nSMase1 (UniProt O60906) · Source: `genes/human/SMPD2/SMPD2-ai-review.yaml`

Executive Judgment

Verdict: Over-annotated (refuted for SMPD2 specifically) for the caveola + mechanical-stimulus pair; core SMase/ceramide function retained.

The caveola (GO:0005901, CC) and response-to-mechanical-stimulus (GO:0009612, BP) annotations on human SMPD2 were propagated by PANTHER phylogenetic inference (IBA/IBD) from an experimental root on rat Smpd2 (Q9ET64, RGD:619753), which in turn rests on Czarny et al. 2003 (P 12473648).

That paper detected a caveolar, flow/pressure-activated neutral sphingomyelinase **immunologically**, using antiserum raised against a purified bovine neutral SMase (Bernardo et al. 2000, P 10713073).

The molecular identity of that bovine antigen was **explicitly not** any then-known SMase ("no apparent homologies ... to any known protein") and was **subsequently cloned as nSMase3 / SMPD4** (Krut et al. 2006,

:

P 16517606

the defining peptide "allowed us to identify ... nSMase3, that shows only a little homology to nSMase1 and -2").

Therefore the caveolar/mechanosensitive activity most plausibly belongs to **SMPD4 (nSMase3)** or an unresolved cross-reactive antigen — **not SMPD2/nSMase1**. No target-specific (RNAi, knockout, tagged-protein, or MS-identified) evidence places SMPD2 itself in caveolae or as a mechanosensor was found. The annotation chain is a textbook paralog/antigen carry-over.

Most important caveats: (1) I could not read the full text of

P 12473648

to independently confirm the seed's claim that its Methods used the Bernardo antiserum and its Discussion links the enzyme to that purified protein — this rests on the seed and on the abstract-level chain, which is nonetheless internally consistent and corroborated by Krut 2006. (2) "Over-annotated" means *mis-attributed to this gene*, not that caveolar nSMase mechanotransduction is unreal — it is real, but the gene identity is the issue. (3) The seed's separate instruction stands: do **not** overturn SMPD2's independent core ceramide-generating function

),

P 25168245

which is molecularly target-specific and out of scope here.

Evidence Matrix

Citation (PMID)	Evidence type	Stance	Claim tested	Key finding	Context
12473648 (Czarny 2003)	Direct assay + localization	Qualifies (source of claim)	Is there a caveolar, mechanoactivated neutral SMase?	Flow/pressure transiently activate a caveola-enriched neutral (not acid) SMase → ceramide → ERK1/2; enzyme detected immunologically, gene identity not molecularly established	Rat lung endothelium in situ; cell-free luminal membranes
15142848 (Czarny & Schnitzer 2004)	Direct assay (pharmacology)	Qualifies	Follow-up mechanotransduction pharmacology	Scyphostatin blocks, C2/C6-ceramide mimics mechanosignaling	Rat lung, bovine aorta EC, cell-free
10713073 (Bernardo 2000)	Direct assay (protein purification)	Refutes SMPD2 identity	What is the purified bovine nSMase antigen?	p46/p97 species; tryptic peptides "showed no apparent homologies ... to any known protein "; peptide antibodies used downstream	Bovine brain
16517606 (Krut 2006)	Structural/ evolutionary + cloning	Refutes SMPD2 identity (competing = SMPD4)	Which gene does the bovine peptide correspond to?	Peptide from the purified bovine nSMase identified nSMase3 (SMPD4) , "only a	Human cDNA; yeast expression

Citation (PMID)	Evidence type	Stance	Claim tested	Key finding	Context
				little homology to nSMase1 and -2"; ER/Golgi, C-tail-anchored, TNF-responsive	
25168245 (Yabu 2015)	Mutant/RNAi + phosphosite	Supports (independent core, out of scope)	Does SMPD2 itself generate ceramide?	JNK phosphorylates nSMase1/SMPD2 at Ser-270; stress/anti-Fas activate it; SMPD2 RNAi lowers ceramide	Zebrafish Z + human Jurkat T
Computational (this run)	Computational	Supports refutation	Are nSMase1/2/3 distinct enough that a peptide couldn't be confused?	Global-alignment pairwise identity ~22% (aligned) among SMPD2/SMPD3/SMPD4	UniProt O60906/Q9NY59/Q9NXE4

GO Curation Implications (leads — require curator verification)

- **GO:0005901 (caveola, CC) on SMPD2** — *Lead: REMOVE or strongly qualify.* The only experimental support (via IBA/IBD from rat Smpd2 IDA) traces to an antibody now known to target nSMase3/SMPD4. SMPD2's own reported localization is ER/nuclear-membrane-associated, not caveolar. If IBA-derived, the correct fix is at the **experimental source annotation** on rat Smpd2

(P 12473648),

which should be re-evaluated; correcting it will re-propagate to human.

- **GO:0009612 (response to mechanical stimulus, BP) on SMPD2** — *Lead: REMOVE or strongly qualify*. Same provenance flaw; mechanosensor role belongs to the caveolar enzyme reagent = SMPD4/nSMase3 lineage, not SMPD2.
 - **Do NOT touch SMPD2's core MF/BP terms for sphingomyelin phosphodiesterase activity / ceramide biosynthesis / apoptotic signaling** — supported by target-specific evidence (P 25168245).
- Avoid a blanket "remove all non-core" that would collateralize the valid function.
- **PANTHER note:** The seed argues PTHR12393 vs PTHR16320 is a family-version difference, not a wrong phylogenetic target. Agreed — the annotation is not defeated by the family-version mismatch. It is defeated by the **flawed experimental root** feeding the IBD/IBA. Curators should treat the PANTHER path as legitimate but the propagated evidence as compromised.

Mechanistic Scope

- **Immediate function under test:** subcellular *localization to caveolae* (CC) and participation in a plasma-membrane *mechanotransduction* response (BP) by the SMPD2 gene product.
 - **Direct gene-product activity actually attributable to SMPD2:** Mg²⁺-dependent neutral sphingomyelinase → ceramide, activated downstream of JNK/stress (P 25168245).
- ER/Golgi/nuclear-membrane associated in the general literature.
- **Downstream/other-gene phenomena mis-attributed:** caveolar sphingomyelin hydrolysis → ceramide → Src/ERK mechanosignaling is a genuine endothelial pathway, but the enzyme is the bovine/nSMase3(SMPD4) lineage, i.e., a *different gene product* performing a similar reaction. Attribution to SMPD2 is inference from a non-isoform-specific reagent, not direct SMPD2 evidence.

Conflicts and Alternatives

1. **Paralog confusion (primary alternative):** the reagent lineage is nSMase3/SMPD4 (Krut 2006). Competing assignment: GO:0005901/GO:0009612 should sit on **SMPD4**, if anywhere, pending isoform-specific validation.
2. **Unresolved / cross-reactive antigen:** Bernardo peptides matched "no known protein"; polyclonal antiserum against a p46/p97 pair could cross-react. The caveolar enzyme could be a mixture or a non-SMPD antigen.
3. **Organism/tissue specificity:** rat lung / bovine aorta endothelium; caveola-rich EC context does not generalize to SMPD2's ER-associated distribution in other cells.
4. **Non-specific pharmacology:** scyphostatin

(P 15142848)

inhibits all Mg^{2+} nSMases, so it cannot discriminate SMPD2 from SMPD3/SMPD4.

5. **Database carry-over:** rat Smpd2 IDA/IMP → PANTHER IBD/IBA → human SMPD2 IBA is a single-donor propagation lacking independent, target-specific confirmation.
-

Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Did P 12473648 Methods/Discussion actually use the Bernardo antiserum and tie the caveolar enzyme to that purified protein?	Abstracts of 12473648/15142848 (no antibody named in abstract); seed asserts full-text usage	Confirms the reagent→antigen link that drives the whole judgment	Read full text of P 12473648 (author-deposited copy) and confirm antibody source and Western identity
Is the rat Smpd2 caveola IDA / mechanical IMP in RGD/QuickGO literally sourced to P 12473648 ?	Seed states QuickGO donor Q9ET64 caveola IDA + mechanical IMP (P 12473648)	Determines whether fixing the source annotation corrects human IBA	Inspect QuickGO/ RGD annotation provenance for Q9ET64
Is there any SMPD2-specific caveolar/mechanosensitive evidence (post-2006)?	PubMed searches returned none tying SMPD2 to caveola/mechanotransduction	Absence would firm up the removal lead	Targeted search + review of SMPD2 KO/tagged-localization studies
Exact SMPD4/nSMase3 caveolar/mechano evidence	Krut 2006 shows ER/Golgi, TNF-responsive; caveola not directly shown for SMPD4	Decides whether the terms should <i>move</i> to SMPD4 vs. be dropped entirely	Isoform-specific localization/mechano assays for SMPD4

Discriminating Tests

- 1. Isoform-specific knockdown/knockout + flow assay:** SMPD2 vs SMPD3 vs SMPD4 siRNA/CRISPR in endothelial cells, then measure flow/pressure-induced neutral-SMase activity and ceramide. Prediction: loss of SMPD4 (not SMPD2) abolishes caveolar mechanoactivation.

2. **Tagged-protein colocalization with caveolin-1** for each isoform under identical conditions; test caveolar residency of SMPD2 directly.
3. **Antibody validation:** confirm the Bernardo/Czarny antiserum reactivity against recombinant SMPD2, SMPD3, SMPD4 (Western) — establishes which isoform(s) it detects.
4. **Reanalysis of caveolar proteomics** (e.g., luminal EC membrane MS) for SMPD2 vs SMPD4 peptides.

Curation Leads (require curator verification)

- **Action change:** Flag GO:0005901 and GO:0009612 on SMPD2 for **removal** / **NOT-qualification**, with the fix ideally applied to the rat Smpd2 experimental source so IBA re-propagation self-corrects.
- **Candidate references to verify (exact snippets):**
 - **P** 16517606

— "a peptide sequence derived from a previously purified bovine neutral sphingomyelinase (nSMase) allowed us to identify a cDNA encoding a novel human sphingomyelinase, nSMase3, that shows only a little homology to nSMase1 and -2." → reassigns reagent to SMPD4.
 - **P** 10713073

— "Amino acid sequencing of tryptic peptides showed no apparent homologies of bovine N-SMase to any known protein." → antigen ≠SMPD2/SMPD3.
 - **P** 12473648

— "primarily in caveolae enriched in sphingomyelin and neutral sphingomyelinase" → origin of caveola/mechano claim, reagent-based.
 - **P** 25168245

— "phosphorylation of nSMase1 (... SMPD2) by ... JNK ... stimulates ceramide generation and apoptosis" → retain core function.
- **Candidate term relocation (optional):** consider whether caveola/mechanotransduction terms belong on **SMPD4/nSMase3** — but only pending direct isoform-specific evidence; do not auto-transfer.

- **Suggested curator questions:** Is the human SMPD2 IBA the only support for these two terms? Does the rat source annotation cite

P 12473648

Is there any post-2006 SMPD2-specific caveolar/mechano paper?

- **Suggested experiments:** the discriminating tests above (isoform-resolved knockdown + flow ceramide assay; antibody cross-reactivity panel).

Iteration 2 addendum — verification attempts & database checks

- **Full text of**

P 12473648

is not machine-accessible. JBC and ScienceDirect returned HTTP 403; Europe PMC reports the article as not open access (no PMCID / full-text XML). I therefore **could not independently confirm** the seed's specific claim that its Methods used preimmune serum + antiserum against the Bernardo bovine nSMase and that the Discussion links the detected enzyme to that purified protein. This remains the key knowledge gap. However, WebSearch of the article confirmed it is antibody/immunoblot-based and does **not** molecularly assign the enzyme to a specific SMPD gene, which is consistent with the abstract-level provenance chain.

- **No SMPD2-specific caveolar/mechanotransduction paper found.** Targeted PubMed queries ("SMPD2 nSMase1 caveolae ... mechanotransduction/shear stress") returned zero hits — supporting the absence of independent, target-specific caveolar/mechanosensor evidence for SMPD2.
- **UniProt curated database check (computed this run):** gene-name mapping confirmed as distinct genes — O60906=**SMPD2**/nSMase1, Q9NY59=**SMPD3**/nSMase2, Q9NXX4=**SMPD4**/nSMase3. UniProt curated subcellular location for **SMPD2** = "**Cell membrane**" (no caveola listed); SMPD3 = Golgi/Cell membrane; SMPD4 = ER/Golgi/nuclear envelope/sarcolemma. So even the reference proteome record does not place SMPD2 in caveolae.
- **Artifacts saved:** `artifacts/evidence_matrix.csv`, `artifacts/go_decision_table.csv` (computed provenance tables).

Iteration 3 note — antibody-specificity chain tightened

Secondary sourcing of Bernardo et al. 2000

(

[P 10713073](#)

clarifies that its **peptide-specific polyclonal antibody was raised against a p97-derived peptide** and **stained a 97-kDa protein in bovine aortic endothelial cell lysates**; the 46-kDa species is a breakdown product, and the antigen is a single protein. This is decisive for two reasons: (1) the p97 species is precisely the sequence that Krut et al. 2006

(

[P 16517606](#)

used to clone **nSMase3/SMPD4**; and (2) the antibody was validated in **bovine aortic endothelial cells**, the same endothelial system used in the Czarny/Schnitzer mechanotransduction work. Thus, to the extent the caveolar mechano-nSMase was identified immunologically with this reagent lineage, it points to **SMPD4/nSMase3**, not SMPD2. The only residual gap is direct confirmation, from the inaccessible

[P 12473648](#)

full text, that this exact antiserum (vs. a generic activity assay) was the identifying reagent in that paper.

Provenance / methods note

Literature retrieved via PubMed (abstracts quoted verbatim). Pairwise identity of human SMPD2/SMPD3/SMPD4 (UniProt O60906/Q9NY59/Q9NXE4) computed in-run by a simple Needleman–Wunsch global alignment; the crude match-only scoring inflates the "% over shorter" figure, so only the aligned-identity (~22%) is interpreted, and solely to corroborate the published "little homology" statement. Full text of

[P 12473648](#)

was not machine-readable in this run; the reagent→antigen link is inferred from the abstract-level chain plus the seed, and is flagged as a knowledge gap.